Pinball Machine

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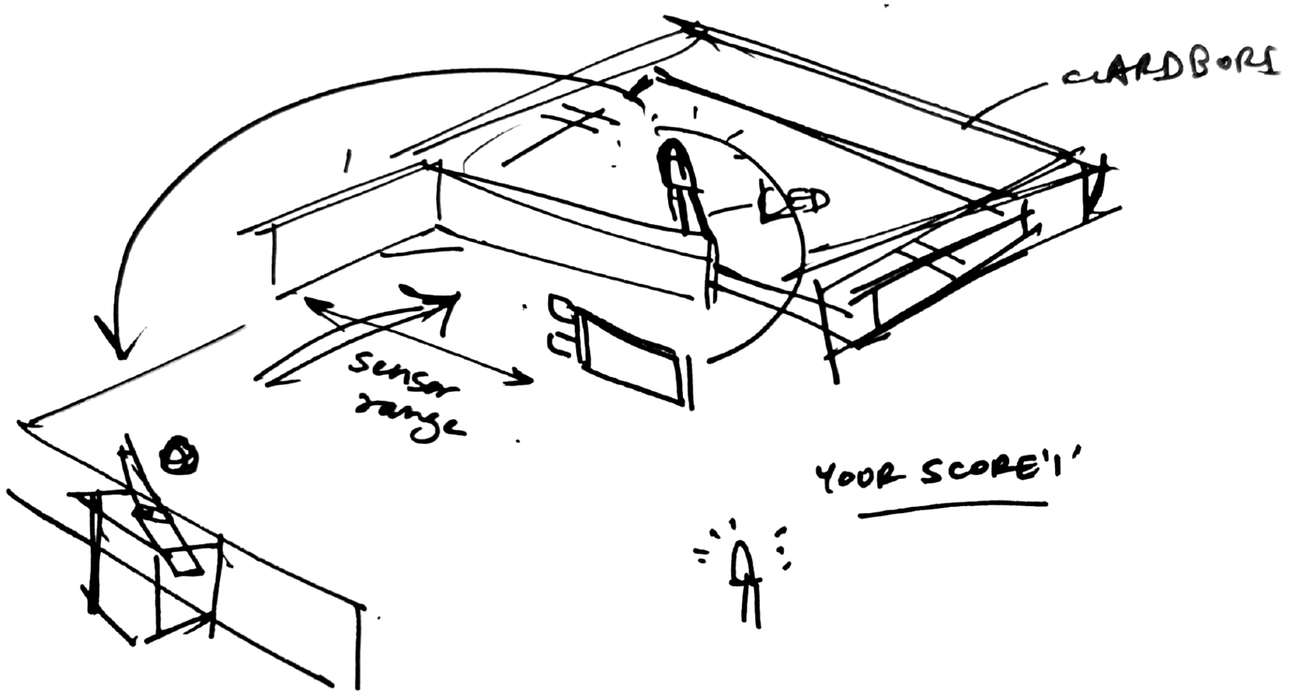
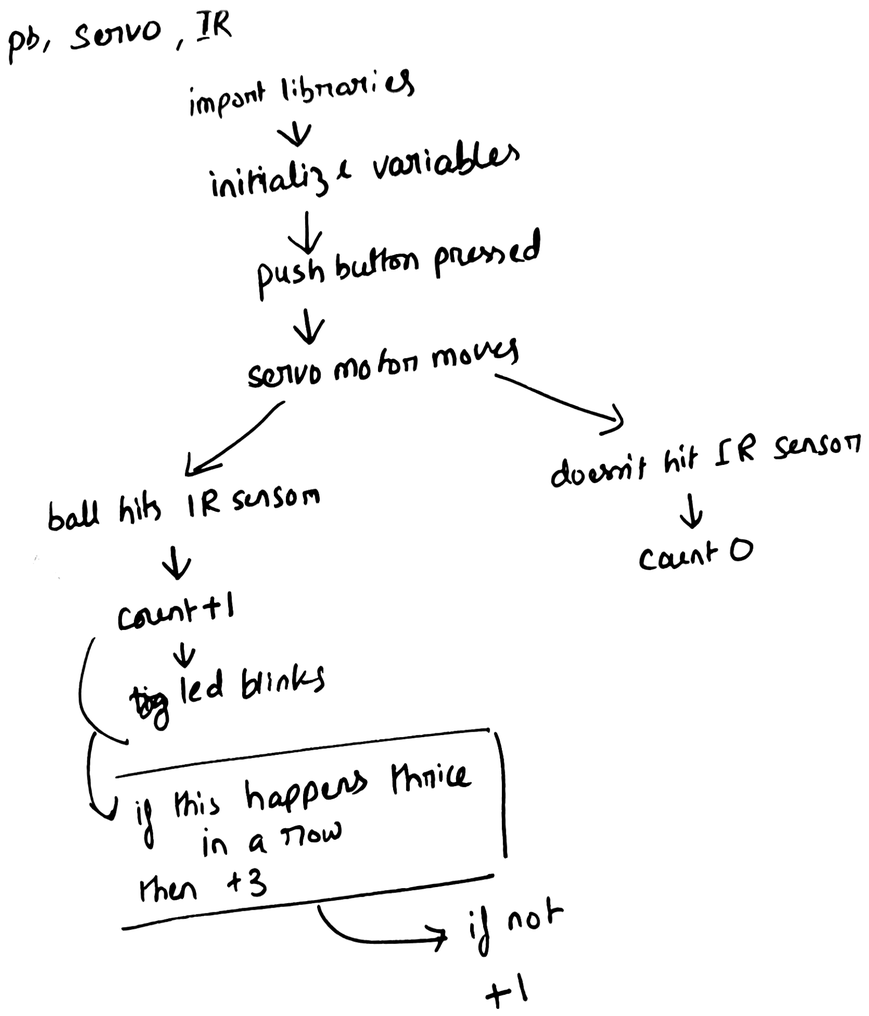
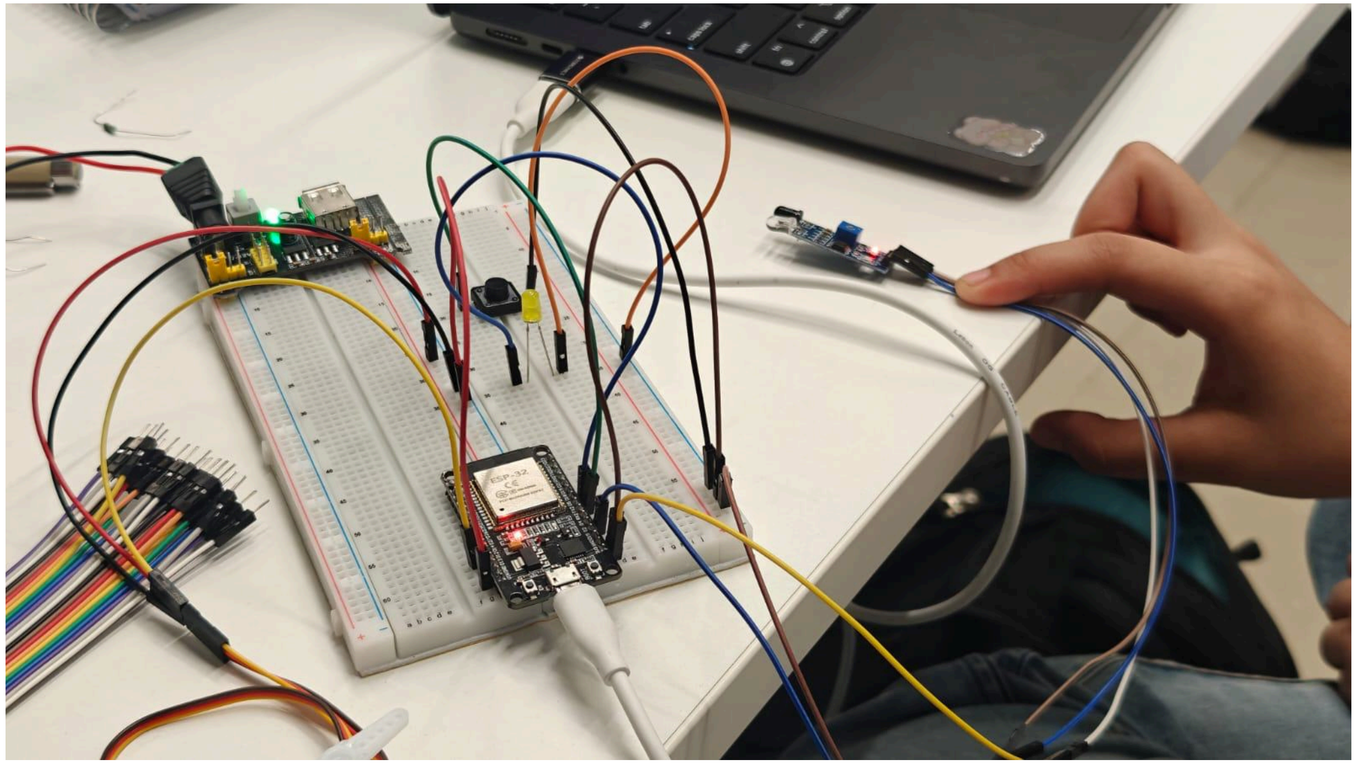
Initial Concept

Pinball machine was one of our first ideas since the module began- A button activates the servo motor which hits a ball, if the ball falls within the range of the IR sensor the score increases and led lights up.
We chose an IR sensor because it works in distant ranges and we cant throw LEDs (LDR sensor)



Initial Flow of Code





```python
from machine import Pin, PWM
import time

#initialize variables
servo=PWM(Pin(14))
sensor=Pin(4,Pin.IN, Pin.PULL_UP)
pb=Pin(5, Pin.IN, Pin.PULL_UP)
led=Pin(18,Pin.OUT)

count=0
servo.freq(50)

print("Welcome to the game!")
time.sleep(0.2)
print("Press the push button to hit the ball!")
while True:
    pb_val=pb.value()
    if pb_val==0:
        for i in range(55,280,50):
            servo.duty(i)
        sensordet=sensor.value()
        if sensordet==0:
            count+=1
            print("Your score is", count)
            time.sleep(0.5)
            led.on()
            time.sleep(0.5)
            led.off()
```

Iteration 1

Works but the IR sensor only increments score when the button is pressed
Another issue was the led was continuously lit up at times

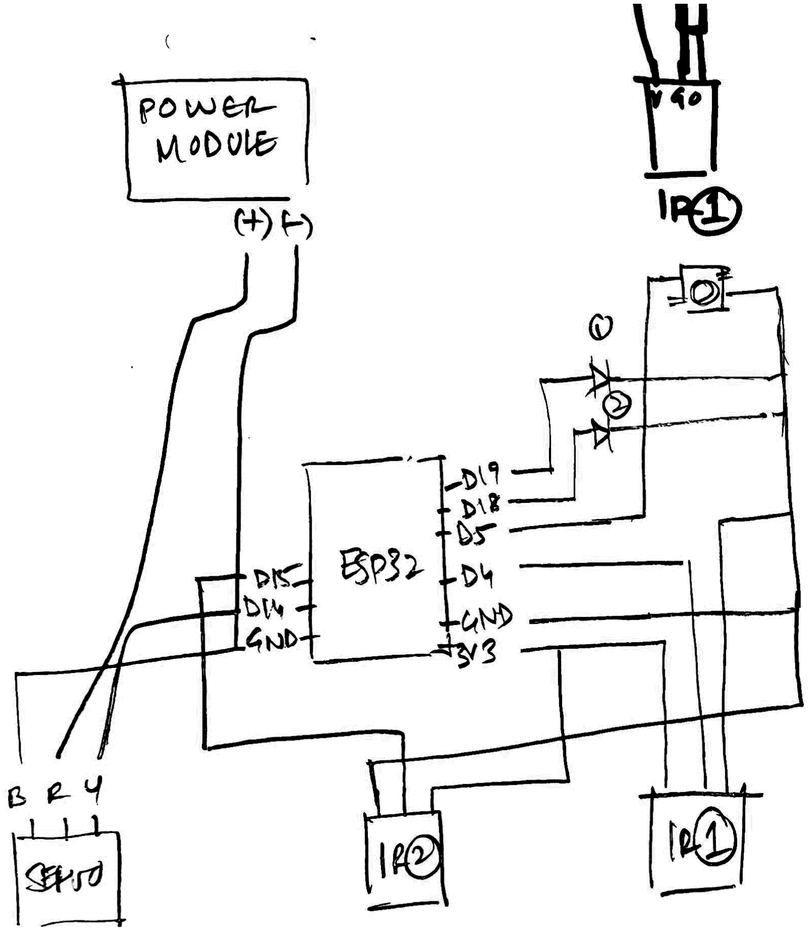
Why is that so

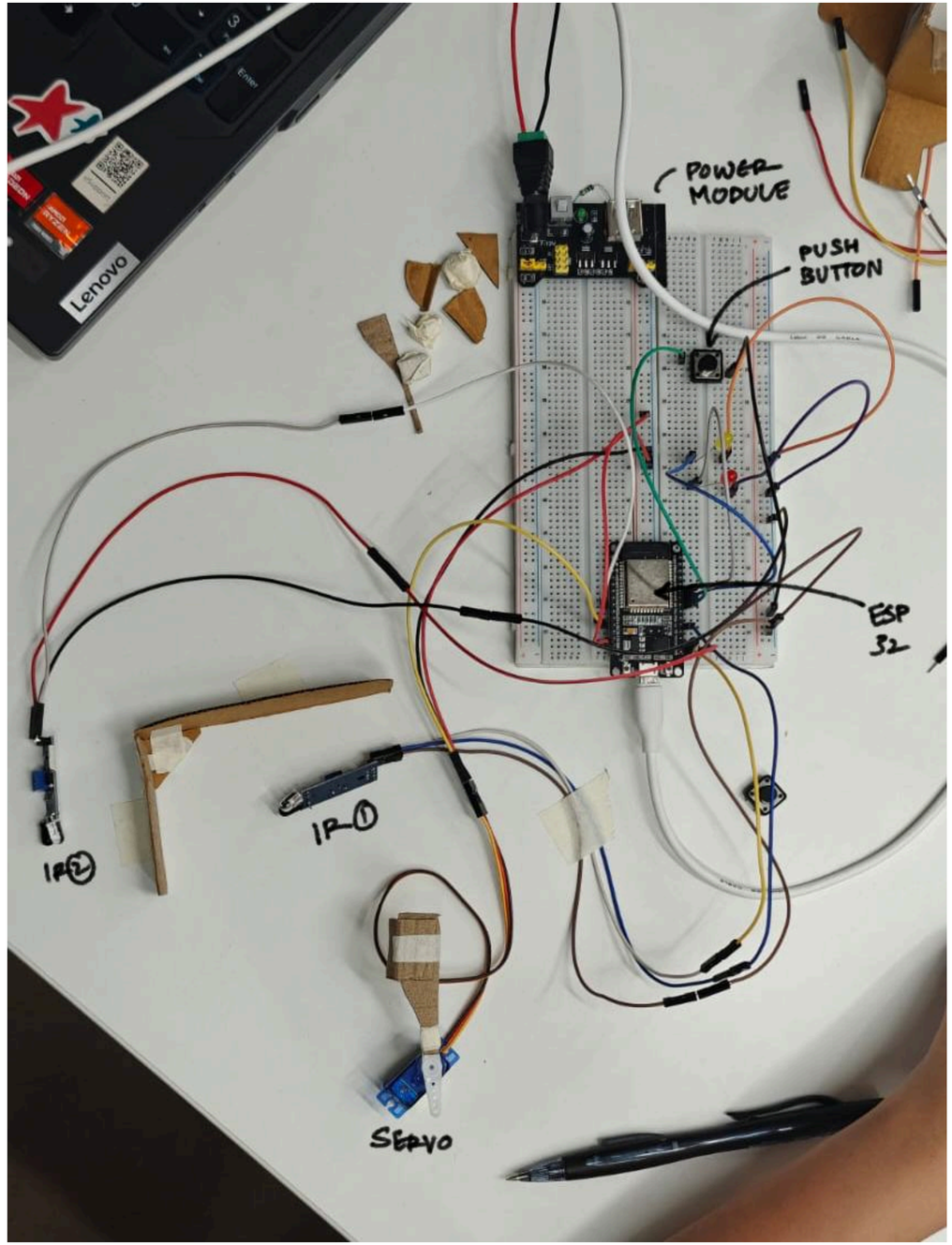
sensordet is under
if pb_val==0:
condition because of that when the value is 1 (ie button not pressed) the code will not execute
if the sensor value becomes 1 before led.off() is read.
To fix this
sensordet was put outside the loop
led.off() was put outside the sensordet if loop

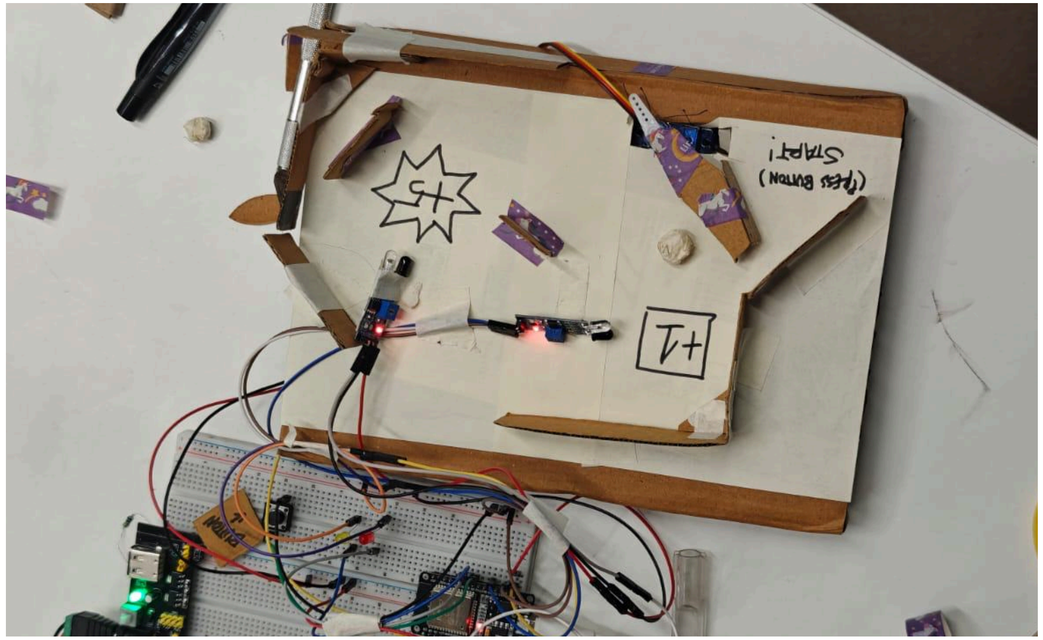
Final Game

We added another IR sensor that would increment +5 when the ball hits it

Components- IR sensor (2), Servo motor, LED (Red and Yellow), Power Module, ESP32, Push Button





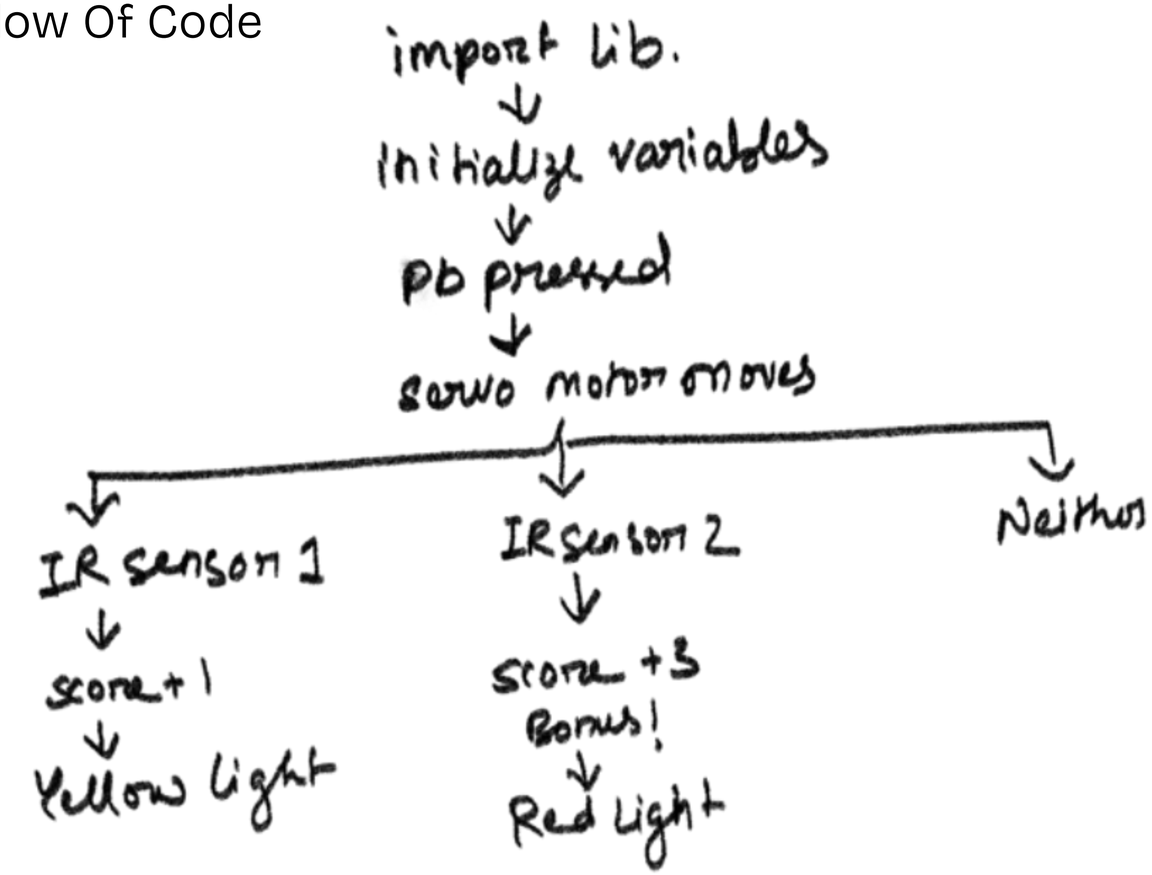


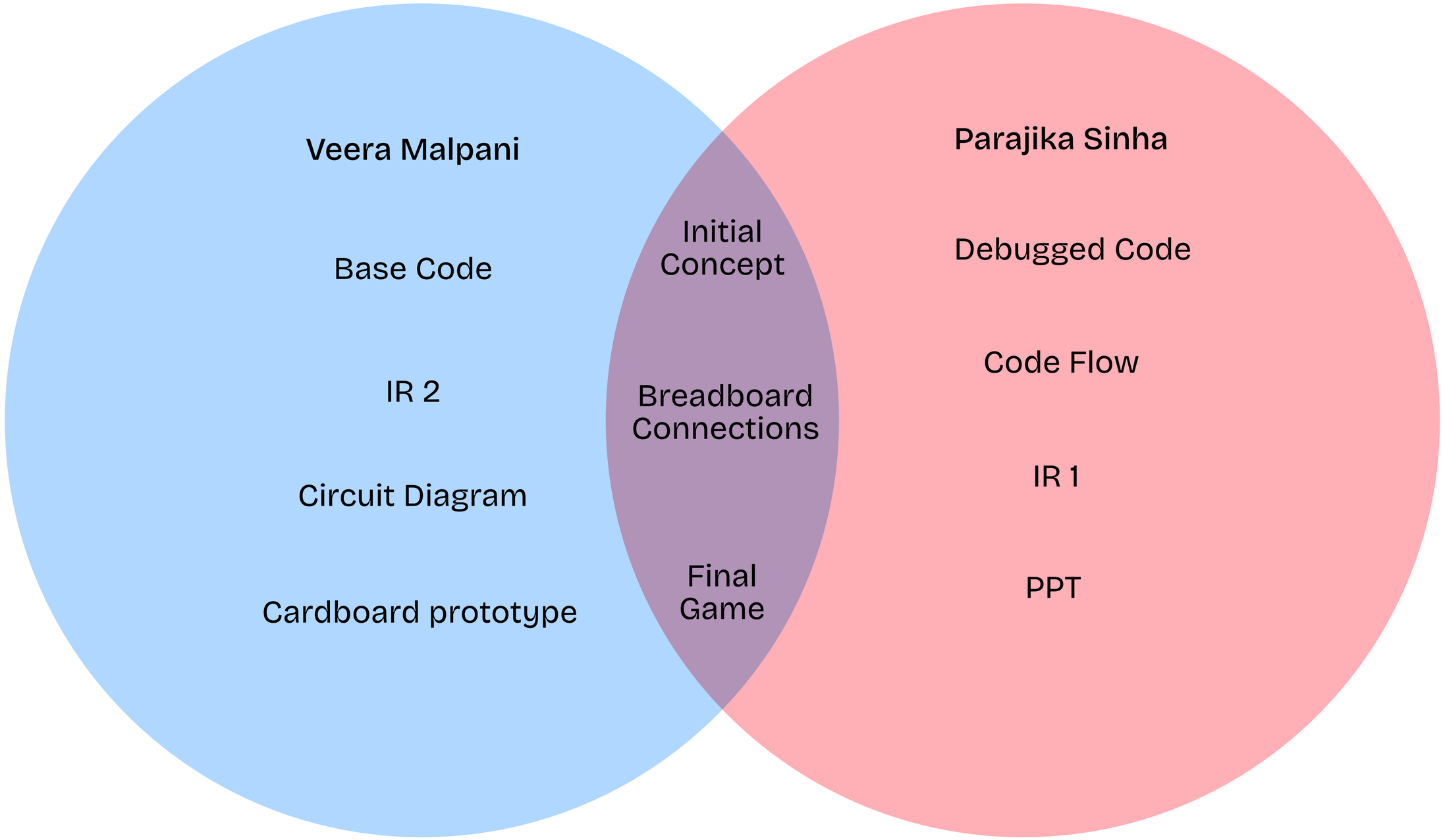
This is an imitation pinball machine, its not perfect and are looking forward improvements but on a base level it has the basic components of what we want
All in all this was a fun project, we learnt more about how the servo motor's duty cycle actually functions

```
MPY: soft reboot
Welcome to the game!
Press the push button to hit the ball!
Bonus! Your score now is 5
Bonus! Your score now is 10
Your score is 11
Your score is 12
Your score is 13
```

The servo motor (moveable) shoots the pinball (angle depending on how long the button is pressed), IR1 adds an increment of 1 to the score qwhere as IR2 adds a bonus score of +5, two obstacles are placed right next to the sensors

Flow Of Code




Veera Malpani
Base Code
IR 2
Circuit Diagram
Cardboard prototype
Initial Concept
Breadboard Connections
Final Game
Parajika Sinha
Debugged Code
Code Flow
IR 1
PPT